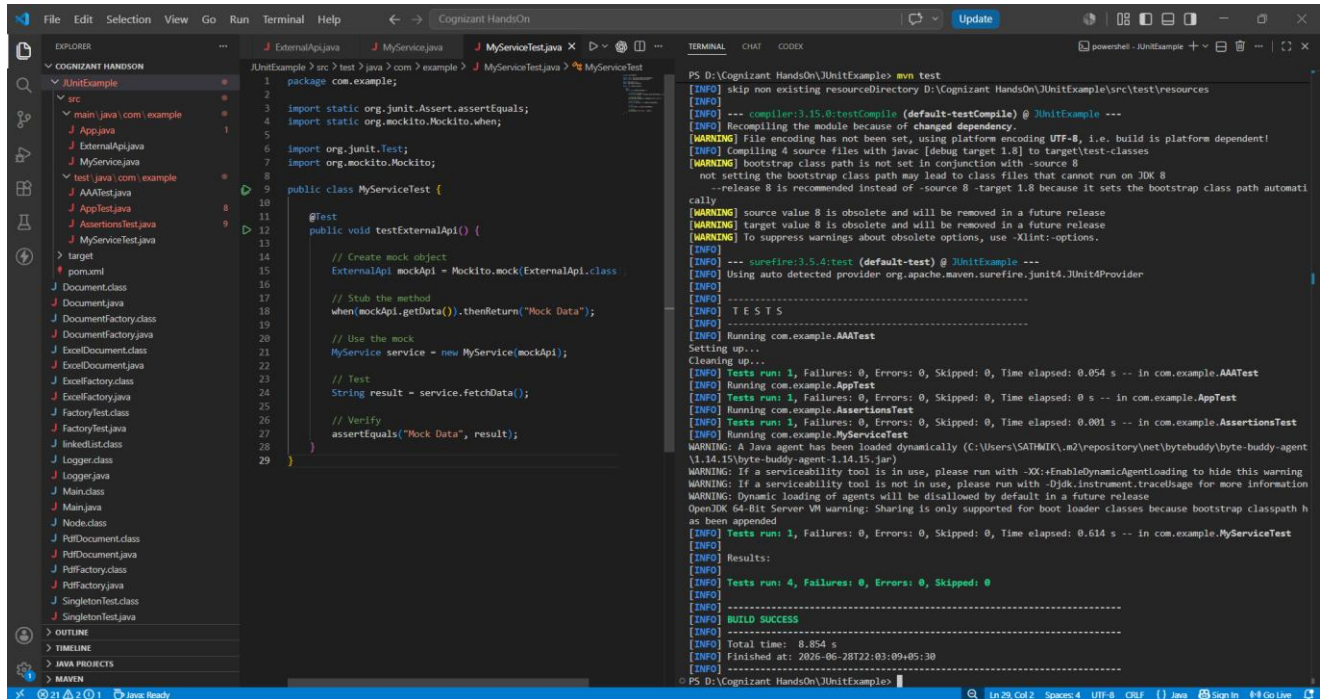


Week 1: Mockito exercises

Exercise 1: Mocking and Stubbing



```
package com.example;

import static org.junit.Assert.assertEquals;
import static org.mockito.Mockito.when;

import org.junit.Test;
import org.mockito.Mockito;

public class MyServiceTest {

    @Test
    public void testExternalApi() {

        // Create mock object
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        // Stub the method
        when(mockApi.getData()).thenReturn("Mock Data");

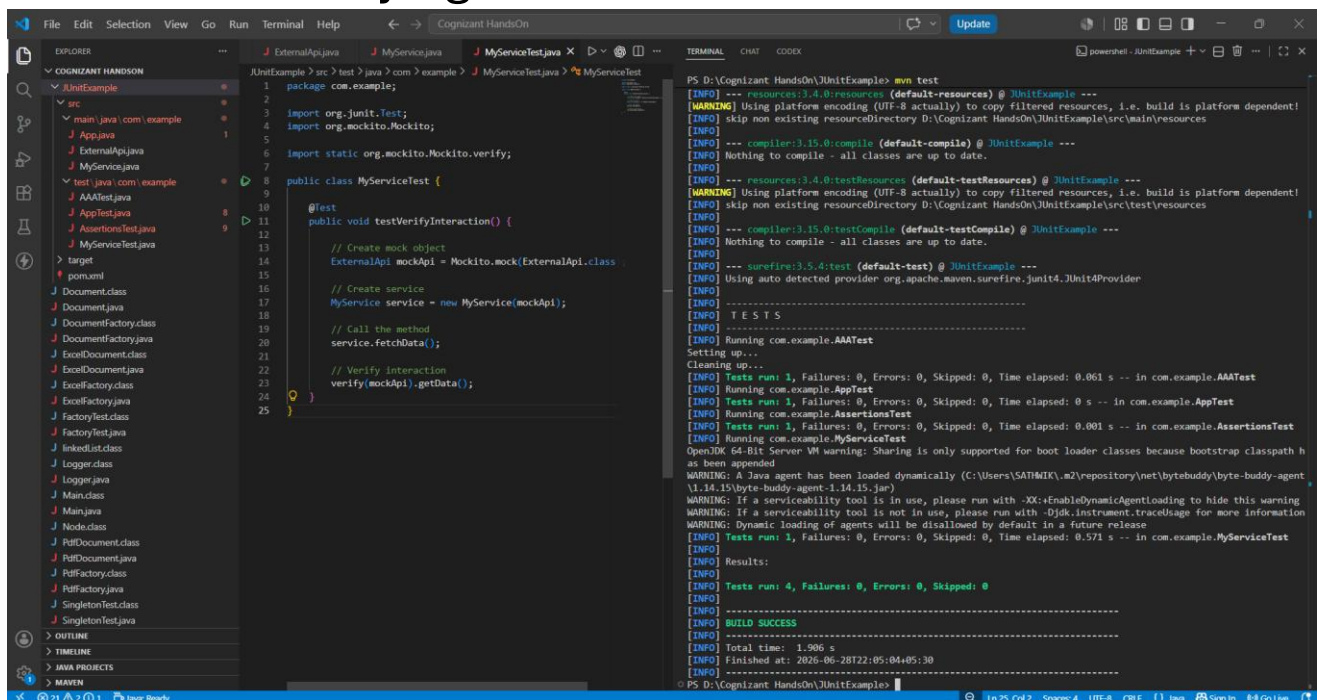
        // Use the mock
        MyService service = new MyService(mockApi);

        // Test
        String result = service.fetchData();

        // Verify
        assertEquals("Mock Data", result);
    }
}
```

```
PS D:\Cognizant HandsOn\JUnitExample> mvn test
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ JUnitExample ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 4 source files with javac [debug target 1.8] to target\test-classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automati
cally
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO] --- surefire:3.5.4:test (default-test) @ JUnitExample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AAATest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.054 s -- in com.example.AAATest
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.AssertionsTest
[INFO] Running com.example.MyServiceTest
[WARNING] A Java agent has been loaded dynamically (C:\Users\SATHIMIK\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar)
[WARNING] If a serviceability tool is in use, please run with -XX:EnableDynamicAgentLoading to hide this warning
[WARNING] If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
[WARNING] Dynamic loading of agents will be disallowed by default in a future release
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath h
as been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 8.854 s
[INFO] Finished at: 2026-06-28T22:03:09+05:30
[INFO] -----
PS D:\Cognizant HandsOn\JUnitExample>
```

Exercise 2: Verifying Interactions



```
package com.example;

import org.junit.Test;
import org.mockito.Mockito;

import static org.mockito.Mockito.verify;

public class MyServiceTest {

    @Test
    public void testVerifyInteraction() {

        // Create mock object
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);

        // Create service
        MyService service = new MyService(mockApi);

        // Call the method
        service.fetchData();

        // Verify interaction
        verify(mockApi).getData();
    }
}
```

```
PS D:\Cognizant HandsOn\JUnitExample> mvn test
[INFO] --- resources:3.4.0:resources (default-resources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ JUnitExample ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\Cognizant HandsOn\JUnitExample\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ JUnitExample ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.5.4:test (default-test) @ JUnitExample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AAATest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.061 s -- in com.example.AAATest
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.AssertionsTest
[INFO] Running com.example.MyServiceTest
[WARNING] A Java agent has been loaded dynamically (C:\Users\SATHIMIK\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar)
[WARNING] If a serviceability tool is in use, please run with -XX:EnableDynamicAgentLoading to hide this warning
[WARNING] If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
[WARNING] Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.071 s -- in com.example.MyServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 1.906 s
[INFO] Finished at: 2026-06-28T22:05:04+05:30
[INFO] -----
PS D:\Cognizant HandsOn\JUnitExample>
```